# Extracting Illicit-Rich Subgraphs from Transaction Networks via Feed-Guided Shortest Paths

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**Abstract.** We study investigative subgraph extraction for Anti-Money Laundering (AML) transaction graphs using the Shortest Paths Network Search Algorithm (SPNSA), which aggregates shortest paths among a small input feed of entities. Since SPNSA is highly sensitive to the feed choice, we evaluate several graph-only feed selection strategies and propose a motif-based coherent feed selection method. Experiments on the IBM-AML benchmark (HI-Small, LI-Small, HI-Medium) and the Elliptic Bitcoin transaction graph show that the proposed strategy yields markedly higher illicit ratios in the extracted subgraph than random and degree-based heuristics on Elliptic and the IBM-AML HI variants, while keeping the output compact.

**Keywords:** anti-money laundering, transaction graphs, investigative subgraph extraction, SPNSA, feed selection, motifs

## 1 Introduction

Money laundering—the process of hiding the illicit origin of funds through complex sequences of transfers—poses an ongoing and evolving challenge for financial institutions. Modern laundering schemes generate multi-step trails naturally represented as directed transaction graphs, where vertices represent entities (e.g., accounts) and directed edges represent transactions. Identifying these patterns is difficult because illicit activity is rare, resulting in highly imbalanced datasets that are difficult to analyze without significant prior information [1,2].

A long-standing barrier in Anti-Money Laundering (AML) research has been the lack of shareable, realistic transaction graphs due to strict privacy and regulatory constraints [7,9]. Recently, Altman et al. [1] addressed this by introducing *AMLworld*, an agent-based simulator, and releasing the IBM-AML benchmark datasets. Complementing these synthetic datasets, the Elliptic Bitcoin dataset provides a real-world transaction graph with labeled illicit/licit nodes [10]. Together, these datasets make it feasible to study graph-based investigative methods on both controlled simulation and real transaction datasets.

Much of recent literature using the IBM-AML benchmark datasets focuses on optimizing detection models (i.e., classifiers) to handle extreme class imbalance and directed multigraph structures [1,5]. For instance, Egressy et al. [5] propose theoretically grounded adaptations that make Graph Neural Networks much more effective on the simulated AML datasets. Blanuša et al. [2] introduce the *Graph Feature Preprocessor* to mine laundering motifs for edge-level features, significantly reducing runtime while maintaining accuracy. Other studies explore privacy-preserving prediction via homomorphic encryption [4] and streaming learning for evolving fraud patterns [3].

In contrast, our work focuses on a task different from supervised detection. We study investigative subgraph extraction, specifically the strategic selection of the algorithm feed. Rather than proposing a new classifier, we aim to produce a compact investigative subgraph with a high proportion of illicit nodes (i.e., high illicit ratio), an essential tool for human analysts who cannot feasibly inspect a global transaction graph. We evaluate this objective on different variants of IBM-AML and Elliptic datasets.

To this end, we build upon the Shortest Paths Network Search Algorithm (SPNSA) proposed by Magalingam et al. [8]. SPNSA takes an input "feed" $F \subseteq V$ (a small set of entities of interest) and constructs a subgraph by aggregating feed-guided shortest paths. For each ego node $s \in F$, SPNSA explores the $r$-hop neighborhood, identifies structurally significant intermediate nodes via centrality measures, and extracts the paths connecting $s$ to these nodes. This approach yields subgraphs significantly more interpretable than naive $k$-hop expansions that often grow rapidly and become difficult to manage in dense networks [8].

However, SPNSA has a critical limitation: its output quality is highly sensitive to the choice of $F$. If the feed nodes are poorly chosen or scattered, the algorithm may route paths through high-connectivity nodes (the financial equivalent of noise) that are not semantically relevant, reducing the illicit ratio of the output.

In addition, the original SPNSA study evaluates the method on the Enron e-mail communication graph from the early 2000s [6,8], which is both domain-specific and temporally distant from modern transaction ecosystems. The study [8] chooses algorithm feeds based on organizational roles in Enron (e.g., selecting top managers) providing an intuitive way to instantiate $F$. In general transaction datasets, such role-based intuition is typically unavailable: we rarely know beforehand which nodes correspond to managers, intermediaries, or suspected launderers; feeds chosen naively could lead SPNSA to extract mostly benign, uninformative subgraphs. This motivates our central research question: *how do we select a feed strategically, using only graph-available signals, such that SPNSA yields a small investigative subgraph enriched with illicit activity?*

We study SPNSA in modern transaction graphs and focus on the problem of selecting the algorithm feed $F$ using only graph-available signals. We evaluate several practical feed selection strategies and systematically analyze how the choice of $F$ affects both the size and illicit enrichment of the extracted SPNSA subgraph across IBM-AML and Elliptic datasets. Based on these findings, we

propose a new motif-based feed selection algorithm that outperforms the other tested strategies and achieves substantial gains on IBM-AML HI variants as well as an approximately 14× higher illicit ratio than random feed selection on Elliptic.

The remainder of the paper is organized as follows. Section 2 describes SPNSA and the feed selection strategies, including our motif-based method. Section 3 presents the datasets, preprocessing, and evaluation protocol. Section 3.5 reports the experimental results, and Section 4 discusses implications and limitations. Finally, Section 5 concludes the paper and outlines future work.

**Code availability.** To support reproducibility, we release the implementation at https://github.com/koshimbetovv/spnsa-feed-selection.

## 2 Methodology

### 2.1 Shortest Paths Network Search Algorithm (SPNSA)

SPNSA is a feed-guided subgraph extraction method designed to produce a compact investigative subgraph when provided with a small, semantically meaningful feed [8]. Given a directed graph $G = (V, E)$ and a feed $F \subseteq V$, SPNSA processes each feed node $s \in F$ as an *ego* node. It first forms the *ego-network* of $s$ within hop radius $r$ (the *SPNSA radius parameter*), i.e., the induced subgraph on nodes whose hop distance (shortest-path length, counting edges as hops) from $s$ is at most $r$. Within this ego-network, SPNSA selects two structurally important *intermediate nodes* using betweenness and eigenvector centrality measures. It then adds the shortest paths (computed inside the ego-network) that connect $s$ to these intermediate nodes and to other feed nodes present in the ego-network [8]. The union of the shortest paths collected over all feed nodes forms the final extracted subgraph, which we call the *shortest path network (SPN)*.

**Algorithm 1** SPNSA (feed-guided shortest paths subgraph extraction), adapted from [8].

**Require:** Graph $G = (V, E)$, feed set $F \subseteq V$, hop radius $r$.
**Ensure:** Extracted subgraph $H = (V_H, E_H)$.
$E_H \leftarrow \emptyset$, $V_H \leftarrow \emptyset$
**for all** $s \in F$ **do**
  $G_s \leftarrow$ ego-network of $s$ in $G$ within $r$ hops
  Select intermediary nodes $M_s \leftarrow$ central nodes in $G_s$ (max betweenness and max eigenvector)
  $T_s \leftarrow (M_s \cup (F \cap V(G_s)))$
  **for all** $t \in T_s$ **do**
    $P \leftarrow$ a shortest path from $s$ to $t$ in $G_s$ (if exists)
    $E_H \leftarrow E_H \cup E(P)$;  $V_H \leftarrow V_H \cup V(P)$
  **end for**
**end for**
**return** $H = G[V_H]$ with edges $E_H$

### 2.2 Feed construction strategies

We compare multiple strategies for constructing a feed $F$ of size $k$. Strategies use only graph topology and simple degree statistics (no illicit labels), except S0 which is a label-informed upper bound used only for reference.

**S0: High illicit degree (label-informed upper bound).** Rank nodes by their number of connections to known illicit nodes and take top-$k$.
**S1: Uniform random.** Sample $k$ nodes uniformly at random from the graph.
**S2: Largest degree difference.** Rank nodes by $|\deg^{out}(v) - \deg^{in}(v)|$ and take the top-$k$ nodes.
**S3: Mixed collect/distribute.** Select two groups of nodes: (i) nodes with $\deg^{in}(v) > 0$ and $\deg^{out}(v) = 0$ (only receivers), taking the top $k/2$ nodes by $\deg^{in}$; and (ii) nodes with $\deg^{out}(v) > 0$ and $\deg^{in}(v) = 0$ (only senders), taking the top $k/2$ by $\deg^{out}$. The final feed set is the union of these two groups.
**S4: Motif-based coherent feed selection.** SPNSA is feed-sensitive: if feed nodes are dispersed, shortest paths can pass through high-connectivity regions, producing large subnetworks with low illicit ratio. Moreover, to reveal illicit structure, the feed should include at least some illicit nodes allowing SPNSA to expand around their neighborhoods and connect them to other suspicious nodes. To address both issues, we propose a motif-based feed selection strategy (S4) targeting nodes likely to participate in laundering patterns visible in graph structure alone, such as short cycles (round-tripping), fan-in/fan-out (aggregation/distribution), and gather–scatter or scatter–gather patterns involving intermediate "relay" vertices linking multiple incoming and outgoing flows [1,2]. Figure 1 illustrates standard transaction motifs commonly used to describe such typologies.

These local patterns suggest that nodes participating in such motifs are more informative feeds than purely random nodes. Accordingly, S4 assigns each node a *motif-based suspicion score* $s(v)$ (see Eqn. 2) using only the local directed neighborhood structure. Nodes with high $s(v)$ are treated as promising feeds. Because computing motif statistics for every node can be expensive at scale, we restrict attention to a candidate pool and compute $s(v)$ only within that pool.

**Candidate pool (scalability).** We construct a candidate pool $\mathcal{C}$ by selecting the top $C$ nodes from the full graph according to a simple scoring rule

$$\deg^{in}(v) \cdot \deg^{out}(v),$$

which prioritizes nodes with both incoming and outgoing activity. We then compute the full motif-based score $s(v)$ only for $v \in \mathcal{C}$.

**Motif scoring.** Let $G = (V, E)$ be the directed transaction graph, where $V$ is the set of accounts and $E \subseteq V \times V$ is the set of directed transaction edges. Let $\Gamma^{in}(v) = \{u : (u \to v) \in E\}$ and $\Gamma^{out}(v) = \{w : (v \to w) \in E\}$. We compute the following motif counts:

$$\begin{aligned}
\mathsf{recip}(v) &:= |\Gamma^{in}(v) \cap \Gamma^{out}(v)| && \text{(reciprocal transfers)},\\
\mathsf{relay}(v) &:= |\Gamma^{in}(v)| \cdot |\Gamma^{out}(v)| && \text{(pass-through/intermediate potential)},\\
\mathsf{cycle3}(v) &:= \sum_{w \in \Gamma^{out}(v)} |\Gamma^{out}(w) \cap \Gamma^{in}(v)| && \text{(directed 3-cycles } u \to v \to w \to u).
\end{aligned}$$

To capture star-like aggregation/distribution while avoiding generic high-degree hubs, we use a thresholded star indicator with star ratio parameter $\tau$, whose dataset-specific values are reported in Table 2:

$$\begin{aligned}
\mathsf{fanin}(v) &:= |\Gamma^{in}(v)| \cdot \mathbb{I}\left[\frac{|\Gamma^{in}(v)|}{|\Gamma^{out}(v)| + 1} \geq \tau\right],\\
\mathsf{fanout}(v) &:= |\Gamma^{out}(v)| \cdot \mathbb{I}\left[\frac{|\Gamma^{out}(v)|}{|\Gamma^{in}(v)| + 1} \geq \tau\right],
\end{aligned} \tag{1}$$

where $\mathbb{I}[\cdot]$ is the indicator function (equal to 1 if the condition holds, and 0 otherwise). Intuitively, $\mathsf{fanin}(v)$ (resp. $\mathsf{fanout}(v)$) equals the in-degree (resp. out-degree) of $v$ only when $v$ is sufficiently imbalanced toward incoming (resp. outgoing) edges by at least a factor $\tau$; otherwise it is set to 0. This helps avoid flagging generic hubs, since nodes with large in- and out-degree typically have ratios close to 1 and therefore fail the $\tau > 1$ threshold, yielding zero score unless their connectivity is strongly one-sided (i.e., star-like).

We then combine these signals with log-compression and an explicit degree penalty to get $s(v)$:

$$\begin{aligned} s(v) \;=\; & \log\big(1+\mathsf{cycle3}(v)\big) + \alpha\,\log\big(1+\mathsf{recip}(v)\big) + \beta\,\log\big(1+\mathsf{relay}(v)\big) \\ & + \mathtt{s_in}\log\big(1+\mathsf{fanin}(v)\big) + \mathtt{s_out}\log\big(1+\mathsf{fanout}(v)\big) \\ & - \gamma\,\log\big(1+\deg(v)\big), \end{aligned} \tag{2}$$

where $\deg(v) = |\Gamma^{in}(v)| + |\Gamma^{out}(v)|$. Here, $\alpha$, $\beta$, $\gamma$, $\mathtt{s_in}$, and $\mathtt{s_out}$ are nonnegative weighting parameters controlling the contribution of reciprocal-transfer, pass-through potential, generic hubs, fan-in, and fan-out terms in the motif score, respectively (Table 2). The log terms prevent a small number of extreme counts from dominating, while the $-\log(1+\deg(v))$ term discourages selecting purely generic hubs that would tend to create "corridor" shortest paths.

**Coherence constraint (multi-center selection).** Let $\pi$ be candidates sorted by decreasing $s(v)$. Rather than taking the top-$k$ nodes globally, we enforce feed coherence by selecting from multiple suspicious regions: we choose the top $m$ nodes in $\pi$ as centers, form the undirected projection $U$ of the graph, and for each center $c$ select high-scoring nodes within the ball $B(c) = \{v : \mathrm{dist}_U(c,v) \le d_{\max}\}$ until the total feed size reaches $k$. Here, $m$, $k$ and $d_{\max}$ are algorithm parameters where $k$ denotes the desired *feed size*, and $d_{\max}$ controls *locality/coherence*.

Using multiple centers reduces the risk of collapsing onto a single region and increases the chance of covering several laundering groups, while local selection increases ego-network overlap and empirically improves illicit enrichment of the SPNSA output. The complete procedure is provided in Algorithm 2.

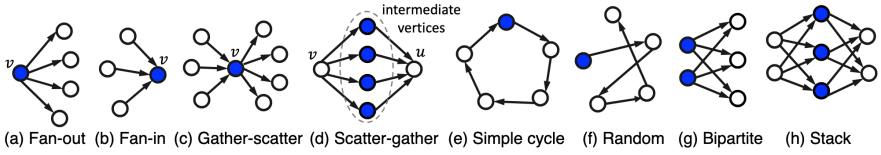


Fig. 1: Local transaction motifs used to model common Money Laundering typologies. Blue nodes indicate key intermediaries in each pattern. Reproduced from [1].

**Algorithm 2** Motif-based Coherent Feed Selection (S4)

**Require:** Directed graph $G = (V, E)$, feed size $k$, candidate pool size $C$, coherence parameters (centers $m$, distance cutoff $d_{\max}$), motif parameters ($\alpha$, $\beta$, $\gamma$ $\tau$, s_in, s_out)
**Ensure:** Feed set $F \subseteq V$ with $|F| = k$

Compute degrees $\deg^{in}(v), \deg^{out}(v)$ for all $v \in V(G)$
Form candidate pool $\mathcal{C}$ by selecting the top-$C$ nodes by a score $\deg^{in}(v) \cdot \deg^{out}(v)$ from $G$
**for** each $v \in \mathcal{C}$ **do**
  $In(v) \leftarrow \{u : (u \to v) \in E(G)\}, \quad Out(v) \leftarrow \{w : (v \to w) \in E(G)\}$
  $\mathsf{recip}(v) \leftarrow |In(v) \cap Out(v)|$ ▷ reciprocal pairs
  $\mathsf{relay}(v) \leftarrow |In(v)| \cdot |Out(v)|$ ▷ pass-through proxy
  $\mathsf{cycle3}(v) \leftarrow \sum_{w \in Out(v)} |Out(w) \cap In(v)|$ ▷ 3-cycles $u \to v \to w \to u$
  $\mathsf{fanin}(v) \leftarrow |In(v)| \cdot \mathbb{I}\big[|In(v)| \geq \tau \cdot (|Out(v)| + 1)\big]$
  $\mathsf{fanout}(v) \leftarrow |Out(v)| \cdot \mathbb{I}\big[|Out(v)| \geq \tau \cdot (|In(v)| + 1)\big]$
  $\deg(v) \leftarrow |In(v)| + |Out(v)|$
  **Score:** $s(v) \leftarrow \log(1 + \mathsf{cycle3}(v)) + \alpha \log(1 + \mathsf{recip}(v)) + \beta \log(1 + \mathsf{relay}(v))$
    $+ \mathsf{s_in} \log(1 + \mathsf{fanin}(v)) + \mathsf{s_out} \log(1 + \mathsf{fanout}(v)) - \gamma \log(1 + \deg(v))$
**end for**
Let $\pi$ be nodes in $\mathcal{C}$ sorted by decreasing $s(v)$
Let $U \leftarrow$ undirected projection of $G$
Initialize $F \leftarrow \emptyset$
Choose centers $A \leftarrow$ first $m$ nodes of $\pi$ (top-scoring centers)
**for** each $c \in A$ **do**
  Compute ball $B(c) \leftarrow \{v \in V(U) : \mathrm{dist}_U(c, v) \leq d_{\max}\}$
  **for** each $v$ in $\pi$ **do**
    **if** $v \in B(c)$ **and** $v \notin F$ **then**
      $F \leftarrow F \cup \{v\}$
      **if** $|F| = k$ **then return** $F$
      **end if**
    **end if**
  **end for**
**end for**
▷ Fallback: fill remaining slots with top-ranked nodes
**for** each $v$ in $\pi$ **do**
  **if** $v \notin F$ **then**
    $F \leftarrow F \cup \{v\}$
    **if** $|F| = k$ **then return** $F$
    **end if**
  **end if**
**end for**
**return** $F$

# 3 Experiments and Results

## 3.1 Datasets

We evaluate our methods on two transaction-graph datasets: the IBM Anti-Money Laundering benchmark (IBM-AML) [1] and the Elliptic Bitcoin transaction dataset [10].

**IBM-AML synthetic benchmark.** The IBM-AML benchmark provides large synthetic AML transaction graphs generated by an agent-based simulator [1]. Accounts are modeled as nodes and transactions as directed, timestamped edges. The benchmark is released at multiple scales (Small/Medium/Large) and activity levels (Higher Illicit ratio, HI / Lower Illicit ratio, LI). In our experiments, we use three IBM-AML variants: HI-Small, LI-Small, and HI-Medium.

**Elliptic Bitcoin transaction dataset.** The Elliptic dataset provides a directed transaction graph derived from Bitcoin transactions, together with illicit/licit labels for a subset of nodes [10]. We use this dataset as a real-world complement to the synthetic IBM-AML benchmark.

### 3.2 Preprocessing

The IBM-AML dataset provides illicit labels at the *transaction (edge)* level [1]. To evaluate illicit enrichment in extracted subgraphs, we derive node labels by marking an account as illicit if it appears as an endpoint of at least one laundering transaction.

In our IBM-AML variants, we observed that 99.6% of illicit-labeled nodes lie in the largest weakly connected component. Therefore, we run all IBM-AML experiments on the largest component to exclude small disconnected components that are not relevant to our evaluation.

In both datasets, we convert the transaction graph to a simple directed graph by keeping a single directed edge $(u, v)$ if at least one transaction from $u$ to $v$ exists. Summary statistics of the preprocessed graphs are reported in Table 1.

Table 1: Dataset statistics used in our experiments. $G$ denotes the (preprocessed) transaction graph; $V(G)$ and $E(G)$ are its vertex and edge sets; $I(G)$ is the set of illicit nodes; and $IR(G) = |I(G)|/|V(G)|$ is the illicit ratio.

| **Dataset** | $\|V(G)\|$ | $\|E(G)\|$ | $\|I(G)\|$ | $IR(G)$ |
|---|---|---|---|---|
| Elliptic | 203769 | 234355 | 4545 | 2.23% |
| HI-Small | 372091 | 874017 | 6342 | 1.70% |
| LI-Small | 504391 | 1184998 | 5285 | 1.04% |
| HI-Medium | 1611244 | 4015935 | 41774 | 2.59% |

### 3.3 Evaluation protocol

For a feed $F$ and SPNSA radius $r$, let $H(F)$ denote the extracted SPN. Let $V(H)$ and $E(H)$ denote its vertex and edge sets, and let $I(H) \subseteq V(H)$ denote the set of illicit nodes in $H$. We report the size of the extracted subgraph as $|V(H)|$ and $|E(H)|$, and we evaluate illicit enrichment using the ratio

$$IR(H) = \frac{|I(H)|}{|V(H)|} = \frac{|\{v \in V(H) : y_v = \text{illicit}\}|}{|V(H)|},$$

where $y_v$ denotes the node label of $v$. Labels are used *only* for evaluation and never for feed construction. For Elliptic, unlabeled nodes are treated as *licit* when computing $IR(H)$.

### 3.4 Experimental settings

We evaluate strategies S0–S4 from Section 2.2. S0 is a label-informed upper bound (it uses illicit labels) and is reported only for reference. We test two feed sizes, $k \in \{200, 500\}$, and two SPNSA radius, $r \in \{1, 2\}$, to capture the trade-off between interpretability (smaller $H$) and coverage (more illicit nodes captured). For S1 (uniform random), we report mean results over 100 random trials.

The hyper-parameters for the motif-based feed selection strategy S4 are reported in Table 2. All remaining strategies are parameter-free given $k$.

Table 2: Hyper-parameters used for strategy S4 across datasets. Parameters are defined in Eqs. (1)–(2) and in Section 2.2.

| Dataset | $C$ | $\alpha$ | $\beta$ | $\gamma$ | $\tau$ | s_in | s_out | $d_{\max}$ | $m$/centers |
|---|---|---|---|---|---|---|---|---|---|
| Elliptic | 300k | 0.2 | 0.3 | 0.8 | 15.0 | 0.5 | 0.4 | 2 | 7 |
| HI-Medium/Small | 300k | 0.2 | 0.3 | 0.8 | 15.0 | 0.3 | 0.4 | 2 | 5 |
| LI-Small | 300k | 0.2 | 0.3 | 0.8 | 27.0 | 0.2 | 0.4 | 2 | 5 |

### 3.5 Results

Table 3 in Appendix A reports the full set of outcomes (subgraph size, number of edges, number of illicit nodes, and illicit ratio) for all feed-selection strategies and SPNSA settings. Because that table is necessarily dense, we summarize the main empirical patterns below and use Figure 2 (illicit ratio comparison) and Figure 3 (size–enrichment trade-off) to highlight what is most practically relevant.

On the Elliptic Bitcoin transaction graph and the two IBM-AML HI variants, the motif-based coherent feed strategy (S4) achieves the highest illicit ratio for every tested SPNSA configuration. Concretely, at $(k, r) = (200, 1)$ on Elliptic, S4 extracts a compact subgraph with only 213 nodes while concentrating illicit activity to an illicit ratio of 33.8% (Table 3(d)). This is far above both the dataset base rate (2.23%, Table 1) and the uniform-random feed baseline (S1), which remains near 2–3% across settings. A similar picture holds on IBM-AML HI-Small and HI-Medium: S4 raises the illicit ratio into the $\approx$ 16%–29% range (depending on $(k, r)$) while keeping the extracted subgraph in the few-hundred-node regime (Table 3(b,c)). Figure 2 makes this ranking visually clear: for Elliptic and both HI graphs, the S4 bar is consistently the highest within each dataset group.

On IBM-AML LI-Small, which has the lowest dataset-level illicit ratio (1.04%, Table 1), the best feed heuristic is the degree-imbalance strategy (S2), not the motif-based strategy. For example, at $(k, r) = (200, 1)$, S2 achieves an illicit ratio of 11.4%, while S4 reaches 5.5% (Table 3(a)). Thus, even though S4 remains clearly better than random, the strongest *structure-only* indicator in this low-illicit setting appears to be captured by simple in/out-degree asymmetry.

Across datasets and strategies, increasing the feed size from $k = 200$ to $k = 500$ predictably increases the extracted subgraph size (more vertices and edges) and often decreases the illicit ratio. A representative example is Elliptic under S4: moving from $(k, r) = (200, 1)$ to $(500, 1)$ increases the extracted subgraph from 213 to 554 nodes, while the illicit ratio drops from 33.8% to 13.0% (Table 3(d)). Increasing the SPNSA radius from $r = 1$ to $r = 2$ similarly enlarges the ego-networks and tends to increase the output size; its effect on illicit ratio is dataset-dependent (e.g., it slightly improves S4 on HI-Medium, Table 3(c)). Overall, these patterns reflect a practical choice: smaller $(k, r)$ typically yields more interpretable, more illicit-concentrated subnetworks, while larger $(k, r)$ increases coverage at the cost of concentration.

Figure 3 jointly considers extracted subgraph size and illicit ratio. For Elliptic and the HI variants, S4 tends to occupy a favorable region of this plane: compared to random feeds (S1) and the simpler degree-based heuristics (S2/S3), it attains higher illicit ratios without requiring larger subnetworks. On LI-Small, the same figure supports the earlier observation that S2 provides the better size–enrichment balance.

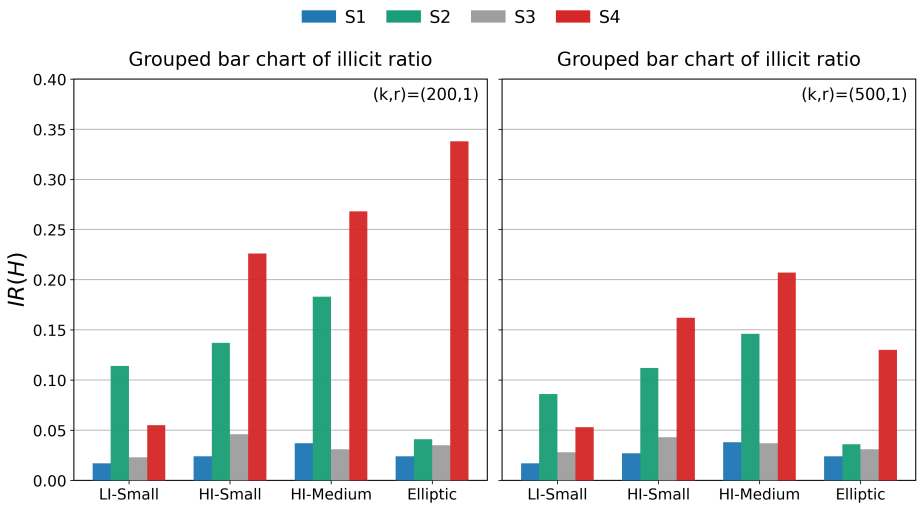


Fig. 2: Grouped bar charts of illicit ratio $IR(H)$ for feed selection strategies S1–S4 on the four datasets, shown for two SPNSA settings: $(k, r) = (200, 1)$ and $(500, 1)$, where $k$ is the feed size and $r$ is the ego-network radius. Each group corresponds to a dataset, and bars compare $IR(H)$ across strategies.

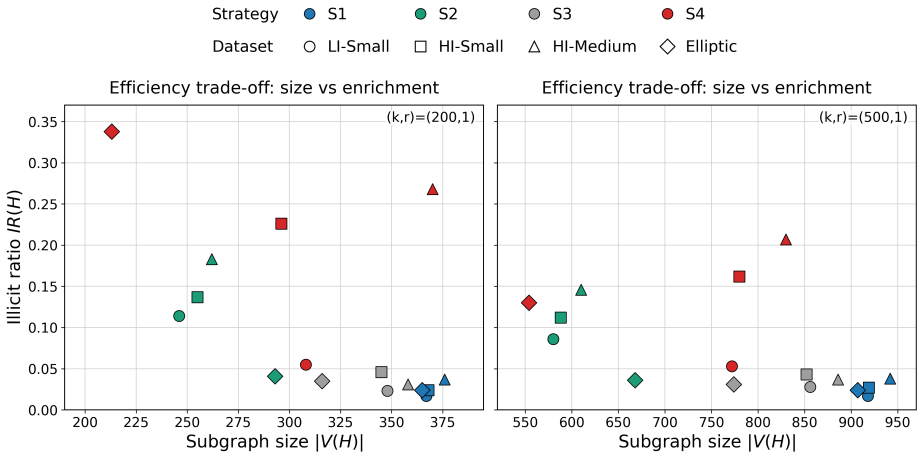


Fig. 3: Efficiency trade-off between extracted subgraph size $|V(H)|$ and illicit ratio $IR(H)$ for feed selection strategies S1–S4 on four datasets, shown for two SPNSA settings: $(k, r) = (200, 1)$ and $(500, 1)$, where $k$ is the feed size and $r$ is the ego-network radius. Colors denote strategies and marker shapes denote datasets; points toward the upper-left indicate a better trade-off.

## 4 Discussion

SPNSA constructs its output by aggregating shortest paths found inside ego-networks around the feed nodes. If the feed is chosen without regard to laundering-relevant structure, these ego-networks are more likely to cover unrelated areas of the graph, and the union of the resulting paths can include many benign vertices, reducing the illicit ratio. The motif-based coherent strategy (S4) targets the opposite behavior: it prioritizes nodes whose local directed neighborhoods are similar to common laundering typologies, and then enforces locality through multi-center selection. Empirically, this combination shifts SPNSA toward extracting subnetworks that are both *compact* and *enriched*, which is exactly the desired outcome for human-centered investigation (Figures 2 and 3).

The LI-Small results show that not every dataset rewards the same structural signal. In that low-illicit dataset setting, the degree-imbalance heuristic (S2) provides the strongest enrichment. (Table 3(a)). One plausible interpretation is that, when illicit activity is sparse, the most informative feed nodes may be those that act as mostly senders or mostly receivers, whereas the specific motif score used by S4 (with the current hyper-parameters) is not as well matched to the dominant illicit structures. This suggests a practical takeaway: feed selection should be treated as a dataset-dependent design choice rather than a fixed preprocessing step.

Across all strategies, larger feeds and larger radius increase the output size and can reduce illicit concentration. For investigative use, the smaller settings (e.g., $k = 200$ and $r = 1$) are attractive starting points. When the goal shifts toward broader coverage, increasing $k$ or $r$ is natural, but the results indicate that one should expect a reduction in illicit ratio.

**Limitations.** First, on Elliptic we treat unlabeled nodes as licit for evaluation, which can underestimate the true illicit ratio. Second, we simplify transaction multi-graphs into simple directed graphs and ignore timestamps during graph construction, removing transaction frequency and temporal information that may be relevant to laundering behavior.

## 5 Conclusion and Future Work

This work studies investigative subgraph extraction in transaction networks through SPNSA, with a focus on a practically central question: how can we select the feed using only graph-available signals, so that SPNSA returns a small investigative subgraph with a high share of illicit activity? We show that this choice strongly changes what SPNSA returns. Across the real Elliptic network and the simulated IBM-AML higher-illicit benchmarks (HI-Small/Medium), our motif-guided coherent feed selection (S4) consistently produces subnetworks with much higher illicit ratios than random feeds and common degree-based heuristics, while keeping the extracted subgraphs small enough to support human inspection. At the same time, the IBM-AML lower-illicit setting (LI-Small) results show that the best graph-only strategy can depend on the dataset regime, and that a simple degree-difference rule can outperform S4 in the low-illicit setting.

Future work includes incorporating temporal and multi-transaction information into both motif scoring and SPNSA aggregation, developing adaptive (data-driven) parameter selection for motif-based feed construction, and validating the findings on additional real transaction graphs as they become available.

# A Appendix

Table 3: SPNSA results under different feed selection strategies for four datasets: IBM-AML LI-Small, HI-Small, HI-Medium, and Elliptic. $H$ denotes the extracted SPN; $V(H)$ and $E(H)$ are its vertex and edge sets, respectively; $I(H)$ is the set of illicit nodes in $H$; and $IR(H) = |I(H)|/|V(H)|$ is the illicit ratio. Here, $k$ denotes the feed size and $r$ denotes the radius of ego-network used in SPNSA. For S1, values are averaged over 100 random trials.

**(a) IBM-AML (LI-Small)**

| | $k$ | $r$ | $\|V\|$ | $\|E\|$ | $\|I\|$ | $IR$ |
|---|---|---|---|---|---|---|
| S0 | 200 | 1 | 449 | 397 | 261 | 0.581 |
| S1 | 200 | 1 | 367 | 168 | 6.4 | 0.017 |
| S2 | 200 | 1 | 246 | 134 | 28 | **0.114** |
| S3 | 200 | 1 | 348 | 148 | 8 | 0.023 |
| S4 | 200 | 1 | 308 | 540 | 17 | 0.055 |
| S0 | 200 | 2 | 612 | 600 | 313 | 0.511 |
| S1 | 200 | 2 | 454 | 257 | 7.8 | 0.017 |
| S2 | 200 | 2 | 309 | 235 | 30 | **0.097** |
| S3 | 200 | 2 | 476 | 276 | 11 | 0.023 |
| S4 | 200 | 2 | 317 | 554 | 15 | 0.047 |
| S0 | 500 | 1 | 1027 | 996 | 594 | 0.578 |
| S1 | 500 | 1 | 918 | 420 | 16.0 | 0.017 |
| S2 | 500 | 1 | 580 | 329 | 50 | **0.086** |
| S3 | 500 | 1 | 856 | 367 | 24 | 0.028 |
| S4 | 500 | 1 | 772 | 1181 | 41 | 0.053 |

**(b) IBM-AML (HI-Small)**

| | $k$ | $r$ | $\|V\|$ | $\|E\|$ | $\|I\|$ | $IR$ |
|---|---|---|---|---|---|---|
| S0 | 200 | 1 | 423 | 394 | 307 | 0.726 |
| S1 | 200 | 1 | 368 | 168 | 8.83 | 0.024 |
| S2 | 200 | 1 | 255 | 134 | 35 | 0.137 |
| S3 | 200 | 1 | 345 | 147 | 16 | 0.046 |
| S4 | 200 | 1 | 296 | 456 | 67 | **0.226** |
| S0 | 200 | 2 | 660 | 778 | 448 | 0.679 |
| S1 | 200 | 2 | 448 | 251 | 13.7 | 0.031 |
| S2 | 200 | 2 | 326 | 249 | 41 | 0.126 |
| S3 | 200 | 2 | 467 | 270 | 21 | 0.045 |
| S4 | 200 | 2 | 313 | 494 | 69 | **0.220** |
| S0 | 500 | 1 | 944 | 960 | 678 | 0.718 |
| S1 | 500 | 1 | 919 | 422 | 24.41 | 0.027 |
| S2 | 500 | 1 | 588 | 320 | 66 | 0.112 |
| S3 | 500 | 1 | 852 | 368 | 37 | 0.043 |
| S4 | 500 | 1 | 780 | 987 | 126 | **0.162** |

**(c) IBM-AML (HI-Medium)**

| | $k$ | $r$ | $\|V\|$ | $\|E\|$ | $\|I\|$ | $IR$ |
|---|---|---|---|---|---|---|
| S0 | 200 | 1 | 431 | 564 | 338 | 0.784 |
| S1 | 200 | 1 | 376 | 176 | 14.07 | 0.037 |
| S2 | 200 | 1 | 262 | 200 | 48 | 0.183 |
| S3 | 200 | 1 | 358 | 158 | 11 | 0.031 |
| S4 | 200 | 1 | 370 | 907 | 99 | **0.268** |
| S0 | 200 | 2 | 756 | 1100 | 590 | 0.780 |
| S1 | 200 | 2 | 462 | 265 | 21.8 | 0.047 |
| S2 | 200 | 2 | 319 | 299 | 63 | 0.198 |
| S3 | 200 | 2 | 459 | 259 | 28 | 0.061 |
| S4 | 200 | 2 | 377 | 973 | 108 | **0.287** |
| S0 | 500 | 1 | 1040 | 1369 | 758 | 0.729 |
| S1 | 500 | 1 | 942 | 442 | 35.5 | 0.038 |
| S2 | 500 | 1 | 610 | 450 | 89 | 0.146 |
| S3 | 500 | 1 | 886 | 386 | 33 | 0.037 |
| S4 | 500 | 1 | 830 | 1623 | 172 | **0.207** |

**(d) Elliptic**

| | $k$ | $r$ | $\|V\|$ | $\|E\|$ | $\|I\|$ | $IR$ |
|---|---|---|---|---|---|---|
| S0 | 200 | 1 | 334 | 191 | 322 | 0.964 |
| S1 | 200 | 1 | 365 | 166 | 8.57 | 0.024 |
| S2 | 200 | 1 | 293 | 140 | 12 | 0.041 |
| S3 | 200 | 1 | 316 | 157 | 11 | 0.035 |
| S4 | 200 | 1 | 213 | 209 | 72 | **0.338** |
| S0 | 200 | 2 | 416 | 309 | 386 | 0.928 |
| S1 | 200 | 2 | 498 | 301 | 10.3 | 0.021 |
| S2 | 200 | 2 | 351 | 232 | 12 | 0.034 |
| S3 | 200 | 2 | 411 | 296 | 12 | 0.029 |
| S4 | 200 | 2 | 218 | 214 | 71 | **0.326** |
| S0 | 500 | 1 | 625 | 498 | 592 | 0.947 |
| S1 | 500 | 1 | 907 | 414 | 21.73 | 0.024 |
| S2 | 500 | 1 | 668 | 372 | 24 | 0.036 |
| S3 | 500 | 1 | 774 | 454 | 24 | 0.031 |
| S4 | 500 | 1 | 554 | 691 | 72 | **0.130** |

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